intuition is that jitters arising from quantum mechanics would be too small to be relevant on astrophysical scales. But with inflation, space expanded at such a colossal rate, doubling in size every $10^{-37}$ seconds, that even a slightly different duration of inflation at nearby locations resulted in a significant wrinkle. In fact, calculations undertaken in specific realizations of inflation have shown that the inhomogeneities produced in this way have a tendency to be too large; researchers often have to adjust details in a given inflationary model (the precise shape of the inflaton field's potential energy bowl) to ensure that the quantum jitters don't predict a universe that's *too* lumpy. And so inflationary cosmology supplies a ready-made mechanism for understanding how the small-scale nonuniformity responsible for lumpy structures like stars and galaxies emerged in a universe that on the largest of scales appears thoroughly homogeneous.

According to inflation, the more than 100 billion galaxies, sparkling throughout space like heavenly diamonds, are nothing but quantum mechanics writ large across the sky. To me, this realization is one of the greatest wonders of the modern scientific age.

## The Golden Age of Cosmology

Dramatic evidence supporting these ideas comes from meticulous satellite-based observations of the microwave background radiation's temperature. I have emphasized a number of times that the temperature of the radiation in one part of the sky agrees with that in another to high accuracy. But what I have yet to mention is that by the fourth digit after the decimal place, the temperatures in different locations *do* differ. Precision measurements, first accomplished in 1992 by COBE (the Cosmic Background Explorer satellite) and more recently by WMAP (the Wilkinson Microwave Anisotropy Probe), have determined that while the temperature might be 2.7249 Kelvin in one spot in space, it might be 2.7250 Kelvin in another, and 2.7251 Kelvin in still another.

The wonderful thing is that these extraordinarily small temperature variations follow a pattern on the sky that can be explained by attributing them to the same mechanism that has been suggested for seeding galaxy formation: quantum fluctuations stretched out by inflation. The rough